

it to the PCB tool

2. Directly make the PCB using the PCB tool

The first approach is more structured, allowing you to avoid many overlooked errors. Make the complete schematic using schematic editor tool 'gshem' (shown in Fig. 1). Attributes for each element, including footprints, should be correct. Otherwise, you will have difficulty importing the schematic to the PCB tool. The schematic editor tool is easy to learn and libraries are very good. You can make any custom symbol of your own too.

Resources supporting this tool are building up very fast. You can check out <http://www.gedasymbols.org/> for symbols and footprint libraries. These custom symbols and footprints can easily be added to the original libraries after simple configurations. For configuration and basic setup of these tools, refer to 'Introducing gEDA' article published in EFY Jan. 2013 issue. Once the schematic is ready, you can immediately import it to the PCB tool. Click 'file' in the PCB tool, followed by 'Import Schematic' (see Fig. 3).

If the footprint attributes are correctly put in the schematic, the schematic will be imported to the PCB tool without any error and you will see something similar to what is shown in Fig. 4.

Now click 'select' and then 'disperse all elements' as shown in Fig. 5. You will see all the elements lined up.

Now place components as you like and press 'o' to optimise the nets. Now you can do the routing using tools on the left side (shown in Fig. 6).

If you want to make the PCB directly, select



Fig. 6: Tools for routing

footprints from the library and join them using rat lines as per the schematic before routing is done. For that, select line from the routing tools and then select rat lines layer. Now join components as per the schematic. Thereafter you can select layers and start routing. If you do not connect components through rat lines, you will not be able to generate the netlist.

Q2. The article 'Getting Started with Raspberry Pi' published in April issue was very interesting and useful. I am done with setting up of the Raspberry Pi and now I want to start programming. Please help me.

Yashoda Gupta
Through e-mail

A2. Programming with Raspberry Pi is easy if you know any of the programming languages supported by it. Raspberry Pi supports Python, C, php and many more. Actually, it is a very wide topic but I would suggest a simple way to start with. Follow the steps below to set up the environment and start programming in the language of your choice:

1. Open the terminal on Raspberry pi and install programming software 'geany' using command:

```
sudo apt-get install python geany
xterm
```

Put the password when prompted.

2. Once installation is complete, run 'geany' software to start programming.

3. You can select the programming language of your choice as shown in Fig. 7. 'geany' will already present a template for that.

4. Write the program in the programming window.

5. Now press the run button indicated by a red box in Fig. 7. The program will now run in a separate window.

6. Fig. 8 shows the output of a simple program in Python.

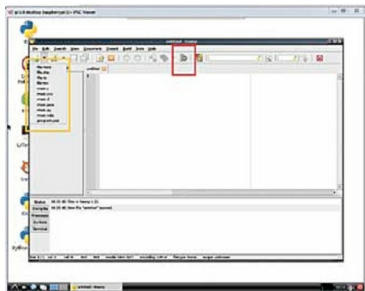


Fig. 7: Programming software 'geany'

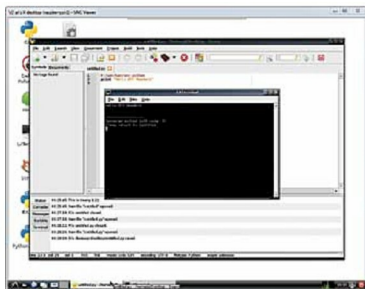


Fig. 8: Output of a simple program in Python

Answers compiled by EFY technical editor, Ankit Gupta. Letters and questions for publication may be addressed to Editor, Electronics For You, D-87/1, Okhla Industrial Area, Phase 1, New Delhi 110020 (E-mail: editsec@efy-india.com) and should include name and address of the sender